

HemoScan AI - Clinical Hematology Report

Clinical Hematology Report

Patient Information:

- * Name: Piyush Kumar Singh
- * Age: 21.0 Years
- * Gender: Male
- * Phone: 9876543210

Complete Blood Count (CBC) Comparison Table

Parameter	Patient Value	Reference Range (Adult Male)	Status
Hemoglobin (Hb)	11.5 g/dL	13.5 – 17.5 g/dL	Low
RBC Count	4.1 x 10 ¹² /μL	4.5 – 5.9 x 10 ¹² /μL	Low
PCV (Hematocrit)	34.2 %	41.0 – 50.0 %	Low
MCV	82.0 fL	80.0 – 100.0 fL	Normal (Borderline)
MCH	26.5 pg	27.0 – 33.0 pg	Low
MCHC	31.5 g/dL	32.0 – 36.0 g/dL	Low

Clinical Interpretation

The laboratory findings for Mr. Piyush Kumar Singh confirm the Machine Learning model's prediction of an Anemic state with a high confidence score of 96.83%.

The patient exhibits a significant reduction in Hemoglobin (11.5 g/dL) and Packed Cell Volume (34.2%), both of which are well below the established physiological requirements for a 21-year-old male. The Total RBC count is also diminished, indicating a reduced oxygen-carrying capacity of the blood.

Analysis of the erythrocyte indices provides deeper insight into the morphology:

- * The MCV (Mean Corpuscular Volume) is at the absolute lower threshold of the normal range (82.0 fL).
- * The MCH (Mean Corpuscular Hemoglobin) and MCHC (Mean Corpuscular Hemoglobin Concentration) are both low, which characterizes the red blood cells as "hypochromic."

This specific combination—where the cells are beginning to shrink (borderline microcytosis) and are deficient in hemoglobin concentration (hypochromia)—is a classic presentation of an evolving nutritional or chronic deficiency. In an adult male, these findings necessitate a search for the underlying cause, as anemia is a symptom of an underlying process rather than a primary diagnosis.

Risk Level: Moderate

The risk is categorized as Moderate. While the hemoglobin level is not currently in the "critical" or "severe" range (typically <7.0 g/dL), a level of 11.5 g/dL in a young male is clinically significant. If left unaddressed, it may lead to chronic fatigue, reduced physical stamina, palpitations, and potential cognitive impairment. The primary concern at this stage is identifying whether the anemia is due to nutritional deficit, occult blood loss, or impaired production.

Possible Anemia Type

Based on the borderline MCV and low MCH/MCHC, the following differentials are considered:

1. Iron Deficiency Anemia (IDA): This is the most probable diagnosis. The indices suggest the early stages of microcytic hypochromic anemia, where the body has exhausted its iron stores and is unable to synthesize

sufficient hemoglobin.

2. Anemia of Chronic Disease (ACD): Often presents as normocytic or borderline microcytic anemia, usually associated with underlying inflammatory or infectious processes.

3. Sideroblastic Anemia or Early Thalassemia Trait: Less likely given the age and specific index values, but remains a secondary differential if iron studies are normal.

Suggested Diagnostic Tests

To confirm the etiology of the anemia, the following investigations are recommended:

1. Iron Profile: Includes Serum Iron, Serum Ferritin (the most sensitive marker for iron stores), Total Iron Binding Capacity (TIBC), and Transferrin Saturation.

2. Peripheral Blood Smear (PBS): A manual microscopic examination to evaluate RBC morphology for signs of anisocytosis (variation in size) and poikilocytosis (variation in shape).

3. Reticulocyte Count: To determine if the bone marrow is responding appropriately by producing new red blood cells.

4. Stool for Occult Blood: To rule out any asymptomatic gastrointestinal bleeding, which is a common cause of iron loss in males.

5. C-Reactive Protein (CRP): To rule out inflammation if Anemia of Chronic Disease is suspected.